THE PUSHFORWARD OF A SHEAF OF SETS UNDER A CONTINUOUS MAP IS A SHEAF

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Sheaf condition on X: sections on an open set are determined by compatible sections on the members of an open covering. invoked at proof lines 2241-2244. Supplies the sheaf property of F that transfers along the covering f^{-1}(V_j).

Lemma 0.2. *Let $f : X \to Y$ be a continuous map. Let $\mathcal{F}$ be a sheaf of sets on X. Then $f_*\mathcal{F}$ is a sheaf on Y.*

Proof. This immediately follows from the fact that if $V = \bigcup V_j$ is an open covering in Y, then $f^{-1}(V) = \bigcup f^{-1}(V_j)$ is an open covering in X. □

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